EC39001: Analog Communication Lab Experiment 8: Pulse Width Modulation (PWM)

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1 Aim of the Experiment

The objectives of this experiment are:

1. To generate a Pulse Width Modulated (PWM) signal using an operational amplifier-based circuit.
2. To study the working of a sawtooth wave generator and comparator in PWM generation.
3. To demodulate the PWM signal using a low pass filter and recover the original message signal.

2 Components Used

- LM 741 Operational Amplifier (used as Integrator)
- IC 710 Comparator
- Resistors: $1k\Omega$, $10k\Omega$, $47k\Omega$, etc.
- Capacitors: $6.8nF$, $100nF$
- Function Generator
- Mixed Signal Oscilloscope (DSO)
- Breadboard and connecting wires
- $\pm 12V$ DC Power Supply

3 Theory & Circuit Diagram

3.1 Pulse Width Modulation (PWM)

Pulse Width Modulation is a technique where the width of the pulses in a pulse train varies according to the instantaneous amplitude of the modulating signal. It is widely used in communication systems and power control applications.

In this experiment, a sawtooth wave acts as the carrier signal, and a sinusoidal wave serves as the modulating signal. The comparator compares both signals, and the output pulse width changes in proportion to the modulating signal amplitude.

3.2 Sawtooth Wave Generator

A sawtooth waveform is generated using a 741 Op-Amp configured as an **integrator**. When a repetitive square pulse is applied to the input, the integrator converts it into a linearly rising and falling voltage waveform (sawtooth).

3.3 Comparator

A comparator (IC 710) compares the instantaneous value of the sawtooth waveform with the modulating signal. When the modulating voltage is greater than the sawtooth voltage, the output is high; otherwise, it is low. This produces the PWM signal.

3.4 Specification

We have used a pulse signal of frequency $f_c = 10kHz$, $and\ Amplitude = 5V\ i.e\quad V_{low} = 0V\ and\ V_{high} = 5V\ and Duty Cycle - 20\%$ .For modulating signal we have used sine wave of frequency=$1kHz$.

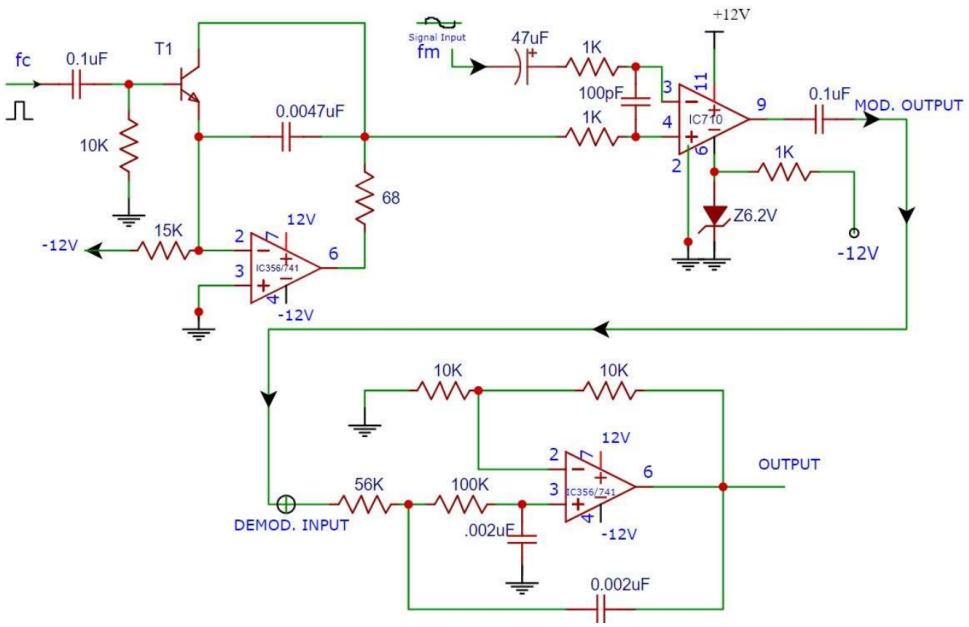


Figure 1: PWM Circuit Diagram

4 Observations and Discussion

4.1 Input and Intermediate Waveforms

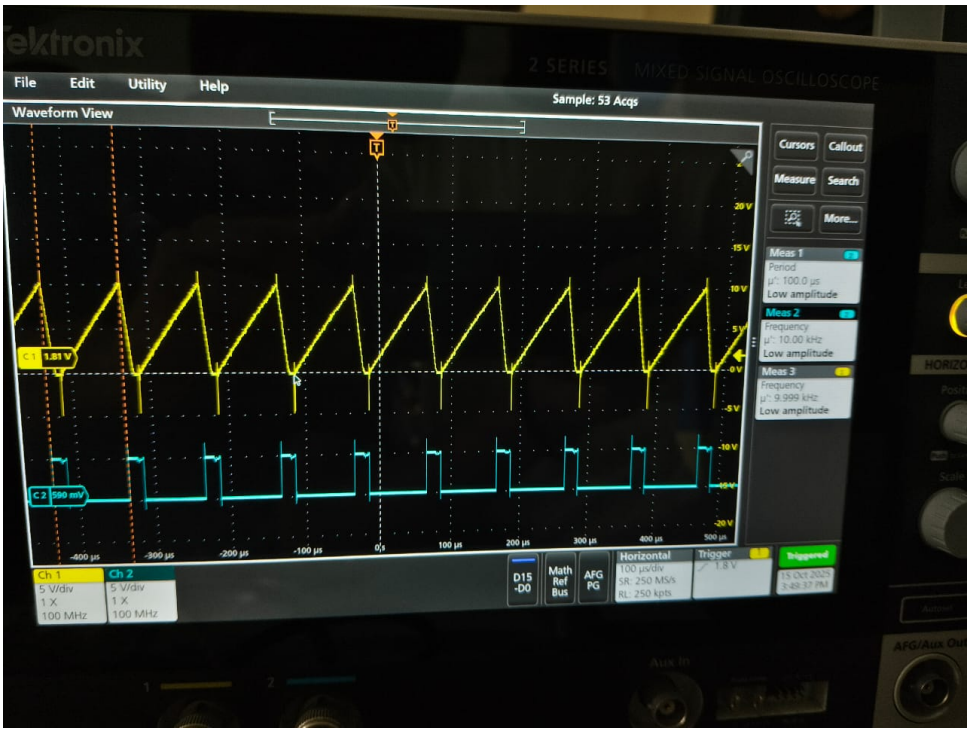


Figure 2: Sawtooth Waveform Output of Integrator and 10 kHz pulse signal

In Figure : 2 we can see the pulse signal (blue) and output of the integrator as the saw tooth wave (yellow) . we can observe that due to loading by the Op-Amp , input signal is bit distored at its peak and the saw tooth wave have little sharp jumps at the minima ,which is possible due to noise introduced by the Op-Amp.

4.2 PWM Output

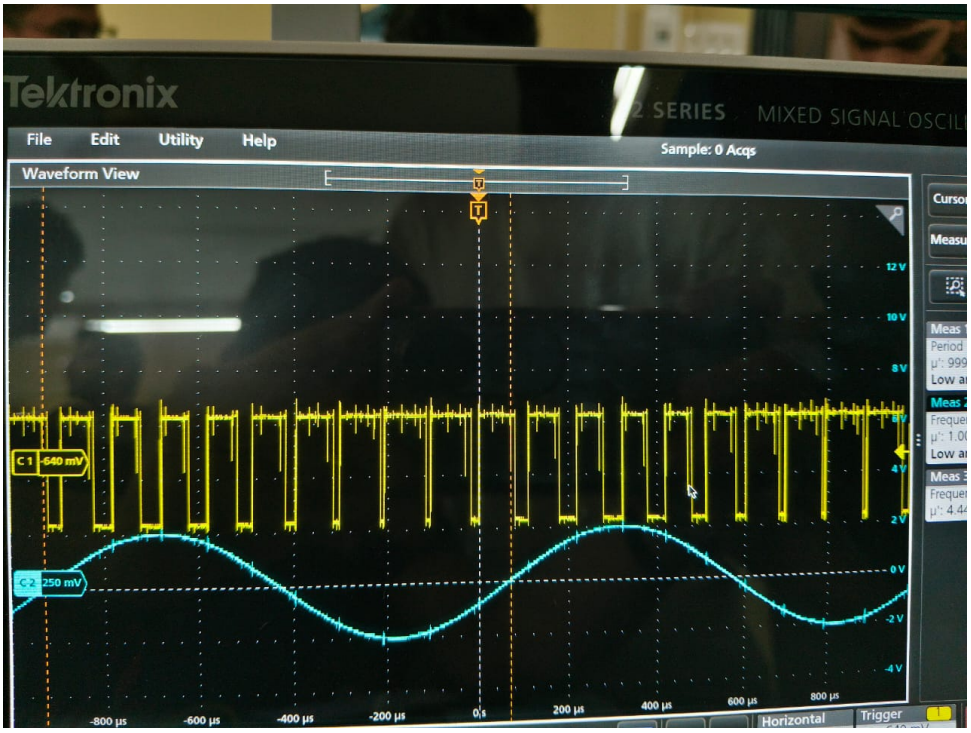


Figure 3: Pulse Width Modulated Output with Modulating signal

The PWM output consists of a series of pulses whose widths change according to the instantaneous amplitude of the modulating signal. In Theory when the modulating signal amplitude is high, pulse width increases, and when it is low, pulse width decreases.But observation says otherwise , as we can observe that there is almost 180° phase shift.



Figure 4: FFT of Modulated Signal

The FFT of modulated signal reveals 1 kHz peak of our modulating signal along with other peaks at multiples of 10 kHz .

4.3 Demodulated Output

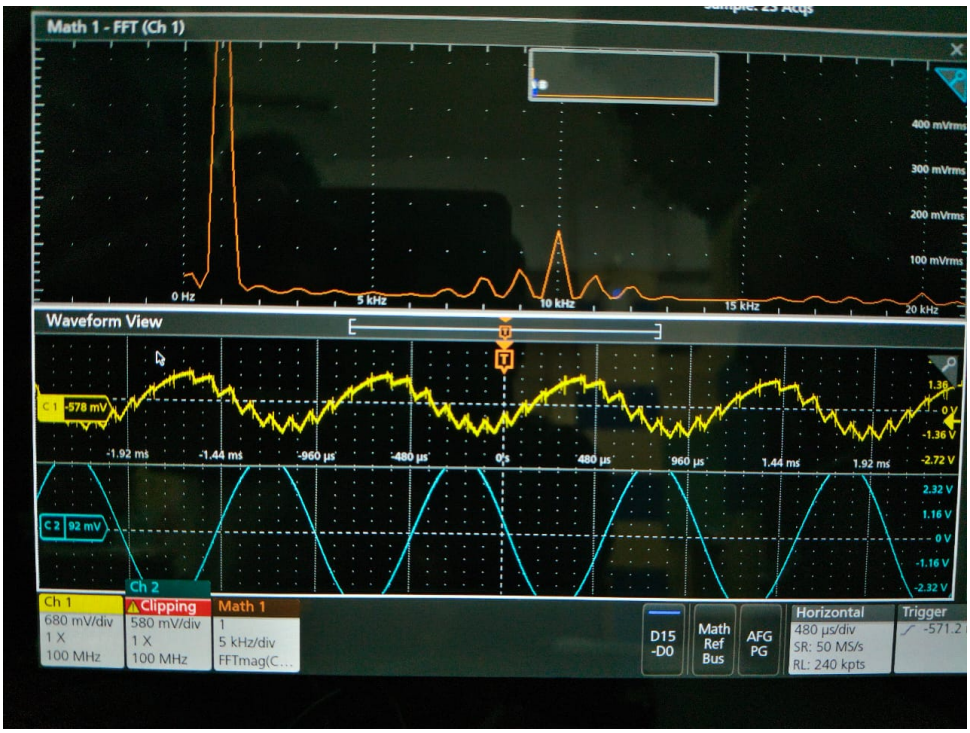


Figure 5: Demodulated Output using Low Pass Filter

The low pass filter removes high-frequency components and reconstructs the original sinusoidal message signal. As observed the signal is noisy with small ripple of sine wave on top of our message signal.It can be verifed by the FFT of the demodulated wave ,which has components other than 1 kHz which is our message signal.

Discussion

- Initially, the sawtooth generator circuit did not produce the expected output because a wrong transistor (PNP BC557) was used instead of the required NPN BC547. Replacing it corrected the waveform generation.
- The sawtooth generator used an op-amp integrator, and the output waveform showed small sharp jumps at the minima due to op-amp noise and limited slew rate.
- The input pulse waveform appeared slightly distorted at its peaks because of loading effects from the op-amp input.
- In Figure 2, the blue waveform represents the input pulse, while the yellow waveform shows the integrator output (sawtooth wave), confirming proper conversion after correction.
- The PWM output consisted of pulses whose widths varied according to the instantaneous amplitude of the modulating signal, but an approximate 180° phase shift was observed compared to the theoretical behavior.
- This phase inversion likely occurred due to the polarity configuration of the comparator inputs.
- The FFT of the modulated PWM signal revealed a dominant component at 1 kHz (modulating signal) along with harmonics at multiples of 10 kHz (carrier frequency), confirming correct modulation.
- The demodulated output obtained using a 2nd-order active low-pass filter (cutoff 1.06 kHz) successfully retrieved the 1 kHz message signal but contained noticeable high-frequency noise.
- The presence of small ripples and unwanted components in the demodulated waveform indicates that the filter did not sufficiently attenuate higher-order harmonics.
- A cleaner output can be achieved by increasing the filter order, which will provide a steeper roll-off and better suppression of unwanted frequency components.

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